

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

PHYSICS — SSC-I (9th Grade) — ANSWER KEY (Version 16 (B))

Syllabus: Unit 1 – Physical Quantities and Measurement, Unit 2 – Kinematics, Unit 3 – Dynamics-I, Unit 4 – Dynamics-II, Unit 5 – Pressure and Deformation, Unit 6 – Work and Energy (to Example 6.4), Unit 9 – Nature of Science and Physics

Total Marks: 38

Time Allowed: 1 Hour 45 Minutes

Version 16 (B)

SECTION A — Answers to Multiple Choice Questions (10 × 1 = 10 Marks)

Q1. A cyclist speeds up uniformly from 5 m/s to 20 m/s in 5 s. His acceleration is:

Correct answer: **A) 3 m/s²** 1 mark

Acceleration is the **gradient** of a speed-time graph: $a = (v - u)/t = (20 - 5)/5 = 15/5 = 3 \text{ m/s}^2$. The *area* under the graph would give distance, not acceleration.

Source in her textbook: Unit 2, §2.5 Acceleration — SLO P-09-B-12. *Retention check on gradient versus area, which she answered correctly on exam-15 after getting it wrong on exam-12 and exam-14.*

Q2. The momentum of a 4 kg body moving at 7 m/s is:

Correct answer: **B) 28 kg m/s** 1 mark

Momentum $p = mv = 4 \times 7 = 28 \text{ kg m/s}$. Note the unit: momentum is measured in kg m/s, not in newtons, which measure force.

Source in her textbook: Unit 3, §3.5 Momentum — SLO P-09-B-38.

Q3. According to Newton's third law, the action and reaction forces are equal and opposite. They act:

Correct answer: **D) on two different bodies** 1 mark

Action and reaction **act on two different bodies**, which is why they never cancel each other out. If both acted on the same body, nothing could ever accelerate.

Source in her textbook: Unit 3, §3.7 Newton's Third Law of Motion — SLO P-09-B-41.

Q4. The single point at which the whole weight of an object appears to act is called its:

Correct answer: **C) centre of gravity** 1 mark

Her textbook defines it in these words: "Centre of gravity is a single point where the whole weight of an object appears to act." The **line of action** is the line along which a force acts, which is a different idea.

Source in her textbook: Unit 4, §4.5 Centre of Mass and Centre of Gravity — SLO P-09-B-49.

Q5. The second condition of equilibrium states that the sum of all torques acting on a body must be:

Correct answer: **A) zero** 1 mark

The second condition of equilibrium is that the resultant torque is **zero**. Her book also calls this the **principle of moments**: anticlockwise torques are taken as positive and clockwise as negative, and the two must balance.

Source in her textbook: Unit 4, §4.4 Conditions of Equilibrium and §4.4.2 Principle of Moments — SLO P-09-B-47.

Q6. The pressure exerted by water at a depth of 4 m is (density of water = 1000 kg/m³, $g = 10 \text{ m/s}^2$):

Correct answer: **C) 40,000 Pa** 1 mark

Liquid pressure $P = \rho gh = 1000 \times 10 \times 4 = 40,000 \text{ Pa}$. Liquid pressure depends only on density, g and **depth**, never on the area of the container's base.

Source in her textbook: Unit 5, §5.4 Pressure in Liquids — SLO P-09-B-82.

Q7. A force of 20 N acts at a perpendicular distance of 0.3 m from a pivot. The torque produced is:

Correct answer: **B) 6 N m** 1 mark

Torque $\tau = F \times L$, where L is the **perpendicular** distance from the pivot: $\tau = 20 \times 0.3 = 6 \text{ N m}$.

Source in her textbook: Unit 4, §4.3 Torque or Moment of Force — SLO P-09-B-44.

Q8. A machine does 1800 J of work in 6 s. Its power is:

Correct answer: **D) 300 W** 1 mark

Power is the rate of doing work: $P = W/t = 1800/6 = 300 \text{ W}$. One watt is one joule per second.

Source in her textbook: Unit 6, §6.4 Power — SLO P-09-B-69.

Q9. A wind turbine generates electricity by converting the:

Correct answer: **A) kinetic energy of moving air into electrical energy** 1 mark

Moving air has **kinetic energy**. Her book states that the rotor blades of a wind turbine capture this energy of the moving air and the generator converts it into electrical energy. Wind is a renewable resource.

Source in her textbook: Unit 6, §6.3 Energy Resources (wind energy) — SLO P-09-B-72.

Q10. The study of the physics of respiration, and of radiation used in diagnosis and treatment, belongs to the interdisciplinary field of:

Correct answer: **B) biophysics** 1 mark

Biophysics applies physics to living systems; her book gives respiration and radiation in diagnostics and treatment as its examples. Astronomy studies objects such as the Sun, Moon, planets and stars.

Source in her textbook: Unit 9, §9.2 Interdisciplinary Fields of Physics — SLO P-09-G-05.

Q	1	2	3	4	5	6	7	8	9	10
Key	A	B	D	C	A	C	B	D	A	B

SECTION B — Answers to Short Questions (3 × 4 = 12 Marks)

Q.2(i) Resultant of 9 N and 12 N at right angles 4 marks

(a) For two perpendicular forces, $F_R = \sqrt{(F_1^2 + F_2^2)} = \sqrt{(9^2 + 12^2)}$ 1

$= \sqrt{(81 + 144)} = \sqrt{225} = 15 \text{ N}$ 1

(b) $\tan\theta = \text{opposite/adjacent} = 12/9 = 1.333$ 1

$\theta = \tan^{-1}(1.333) = 53.1^\circ$ from the 9 N force (about 53°) 1

Marking note: accept 53° to 53.2° . If she measures the angle from the 12 N force instead and says so, that is 36.9° and is also acceptable, since the question's reference force is then stated.

Source in her textbook: Unit 1, §1.8 Addition of Vectors — SLO P-09-A-10 (determine, by calculation or graphically, the resultant of two vectors at right angles).

Q.2(ii) Metre rule balanced on a central pivot 4 marks

(a) By the principle of moments, anticlockwise torque = clockwise torque 1

$6 \text{ N} \times 20 \text{ cm} = W \times 15 \text{ cm}$, that is $6 \times 0.20 = W \times 0.15$ 1

$1.2 = 0.15 W$, so $W = 1.2 / 0.15 = 8 \text{ N}$ 1

(b) The **second condition of equilibrium** (the principle of moments): the sum of the torques is zero. 1

Marking note: full credit if she works in centimetres throughout ($6 \times 20 = W \times 15$), since the units cancel. The (b) mark needs the condition **named**, not just used.

Source in her textbook: Unit 4, §4.4 Conditions of Equilibrium, §4.4.2 Principle of Moments — SLOs P-09-B-46 and P-09-B-47.

Q.2(iii) Work and power climbing a staircase 4 marks

(a) Work against gravity $W = mgh = 50 \times 10 \times 4$ 1 = 2000 J 1

(b) Power $P = W/t = 2000/8$ 1 = 250 W 1

Marking note: the unit must appear on both answers. J and W, not just the numbers.

Source in her textbook: Unit 6, §6.1 Work, §6.2 Potential Energy and §6.4 Power — SLOs P-09-B-61, P-09-B-65 and P-09-B-69.

SECTION C — Answers to Long Questions (2 × 8 = 16 Marks)

Q.3(i) Car of mass 600 kg, rest to 20 m/s in 10 s 8 marks

(a) $a = (v - u)/t = (20 - 0)/10$ 1 = **2 m/s²** 1

(b) $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 2 \times 10^2$ 1 = **100 m** 1

Accept also $s = ((u + v)/2) \times t = (20/2) \times 10 = 100$ m.

(c) $F = ma = 600 \times 2$ 1 = **1200 N** 1

(d) Change in momentum $\Delta p = mv - mu = (600 \times 20) - 0$ 1 = **12,000 kg m/s**, and the unit is **kg m/s** (also written N s) 1

Marking note: (d) explicitly asks for the unit, so the second mark depends on it being written.

Source in her textbook: Unit 2, §2.5 and the equations of motion; Unit 3, §3.4 Newton's Second Law and §3.5 Momentum — SLOs P-09-B-10, P-09-B-36 and P-09-B-38.

Q.3(ii) Hydraulic jack 8 marks

(a) $P = F/A = 100 / 0.004$ 1 = **25,000 Pa** (2.5×10^4 Pa) 1

(b) By Pascal's principle the same pressure acts on the large piston, so $F = P \times A = 25,000 \times 0.2$ 1 = **5000 N** 1

(c) Force ratio = $5000/100 = 50$; area ratio = $0.2/0.004 = 50$. The two ratios are equal, so the force is multiplied by exactly the ratio of the areas. 2

(d) A liquid is **almost incompressible**, so the pressure applied at the small piston is transmitted undiminished to the large piston and the pistons move at once. A gas would compress instead, absorbing the effort, and the large piston would barely move. 2

Source in her textbook: Unit 5, §5.6 Pascal's Principle and §5.7.1 Hydraulic Lift — SLOs P-09-B-86 and P-09-B-87.

— END OF ANSWER KEY —